

Microprocessors & Microcontrollers

BRANCH: ELECTRONICS (ECE)

COURSE CODE: EC301

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Microprocessors & Microcontrollers (EC301) under the Electronics (ECE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: 8085 Microprocessor

8085 Architecture, Register organization, Pin diagram, ALU. Instruction set: Data transfer, Arithmetic, Logical, Branch control instructions. Assembly Language Programming.

Unit 2: 8086 Microprocessor

8086 Architecture: BIU, EU. Memory segmentation, physical address calculation. Minimum and Maximum mode operations. Interrupt structure.

Unit 3: Interfacing Devices

Interfacing memory and I/O devices. Programmable Peripheral Interface (8255), Programmable Interrupt Controller (8259), Keyboard/Display controller (8279), USART (8251).

Unit 4: 8051 Microcontroller

Introduction to Microcontrollers, 8051 block diagram, architecture, memory map, SFRs, Timers, Counters, Serial communication, and basic interfacing.

*8086 Physical Address = Segment Address * 16 + Offset Address. 8085 has 5 hardware interrupts (TRAP, RST7.5, RST6.5, RST5.5, INT0).*

Key Formulas & Exam Pointers



